Computer Network And Security

Unit II: WIRELESS STANDARDS

- **Wireless LANs (WLAN):**
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  - Design goals and Characteristics
  - Network Architecture
- **IEEE 802.11:**
  - Components in IEEE 802.11 network
  - Physical Layer
  - MAC Sub Layers (DCF, PCF)
  - Hidden and Exposed Station Problem
  - Frame Format
  - Addressing Mechanism
- **IEEE 802.15.1 Bluetooth:**
  - Architecture
  - Layers
  - Operational States
- **IEEE 802.16 WiMax:**
  - Services
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  - Layers
- **Comparison between Bluetooth, IEEE 802.11, and IEEE 802.16**

1. Fundamentals of WLAN

Wireless Local Area Networks (WLANs) allow devices to connect and communicate wirelessly within a limited area, such as a home, school, or office building.

- **Medium**: Unlike wired LANs (Ethernet), WLANs use Radio Waves (RF) or Infrared signals for transmission.
- **Standards**: Primarily governed by the IEEE 802.11 family of standards (Wi-Fi).
- **Flexibility**: Provides "anywhere, anytime" connectivity within the range of an Access Point (AP).

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2. Design Goals and Characteristics

In an exam, listing these as bullet points demonstrates a clear understanding of the engineering constraints of wireless tech.

Design Goals:

- **Global Operation:** Must operate in unlicensed frequency bands (like 2.4GHz or 5GHz) available worldwide.
- **Low Power:** Devices often run on batteries; protocols must minimize power consumption.
- **Easy to Install:** Should not require complex cabling or professional infrastructure setup.
- **Robustness**: Must handle interference from microwave ovens, cordless phones, and other electronic noise.

****Characteristics:**

- **Mobility**: Supports seamless connectivity while the user is moving.
- **Scalability**: Can easily add more users by adding more Access Points.
- **Propagations Issues**: Signals are subject to fading, multi-path propagation, and shadowing.
- **Shared Medium**: All devices share the same airwaves, meaning collisions can occur (managed by CSMA/CA).

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3. Network Architecture

This is the most technical part of the answer and carries the most weight. You must explain the two primary modes of operation.

A. Infrastructure Mode (BSS & ESS)

This is the most common setup where devices communicate through a central hub.

- **BSS** (Basic Service Set): A single Access Point (AP) and all the wireless stations (STAs) connected to it.
- **ESS** (Extended Service Set): Created by connecting multiple BSSs via a Distribution System (DS), usually a wired Ethernet backbone. This allows for "roaming."

B. Ad-hoc Mode (IBSS)

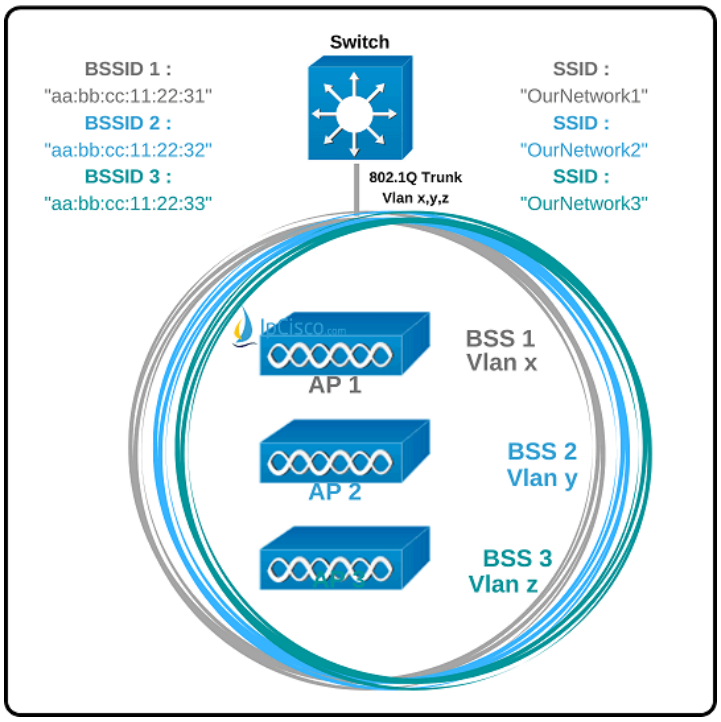
- IBSS (Independent Basic Service Set): A peer-to-peer network where stations communicate directly with each other without an Access Point.
- Use Case: Quick file transfers between two laptops or emergency setups where no infrastructure exists.

Key Components to Mention:

1. **Station (STA)**: Any device with a wireless interface (Laptop, Phone).
2. **Access Point (AP)**: Acts as a bridge between the wireless and wired world.
3. **Distribution System (DS):** The logical component used to interconnect BSSs.

Comparison: BSS vs. ESS

Feature	Basic Service Set (BSS)	Extended Service Set (ESS)
Definition	A single Access Point (AP) with its associated stations (STAs).	A set of two or more BSSs interconnected together.
Core Components	1 Access Point + Wireless Clients.	Multiple APs + **Distribution System (DS)**.
Infrastructure	Works in **Infrastructure mode** or **Ad-hoc mode**.	Always requires a **Wired/Wireless Backbone** (DS).
Roaming	Not supported (connection drops if you move to a new AP).	**Seamless Roaming** supported between different APs.
Range/Size	Small (limited to the range of one AP).	Large (can cover entire buildings or campuses).
Typical Use	Homes or small cafes.	Universities, Hospitals, or Large Offices.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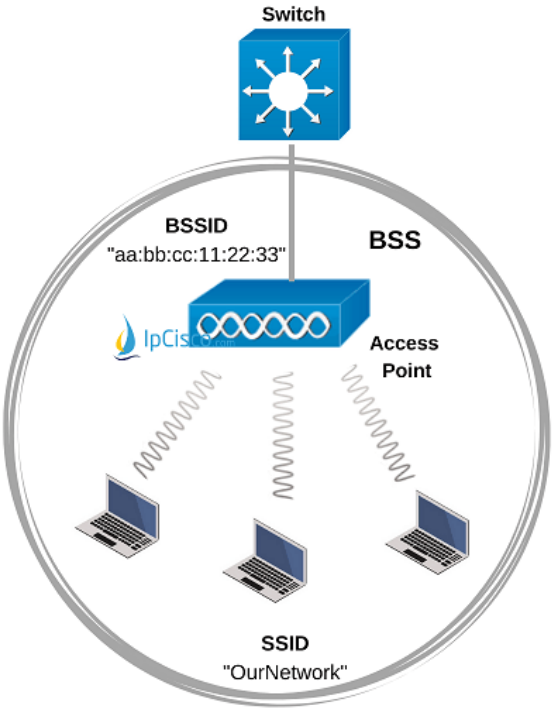
"aa:bb:cc:11:22:31"
BSSID 2 :
"aa:bb:cc:11:22:32"
BSSID 3 :
"aa:bb:cc:11:22:33"
802.1Q Trunk
Vlan x,y,z
SSID :
"OurNetwork1"
SSID :
"OurNetwork2"
SSID :
"OurNetwork3"
AP 1
BSS 1
Vlan x
AP 2
BSS 2
Vlan y
BSS 3
Vlan z


Switch
BSSID
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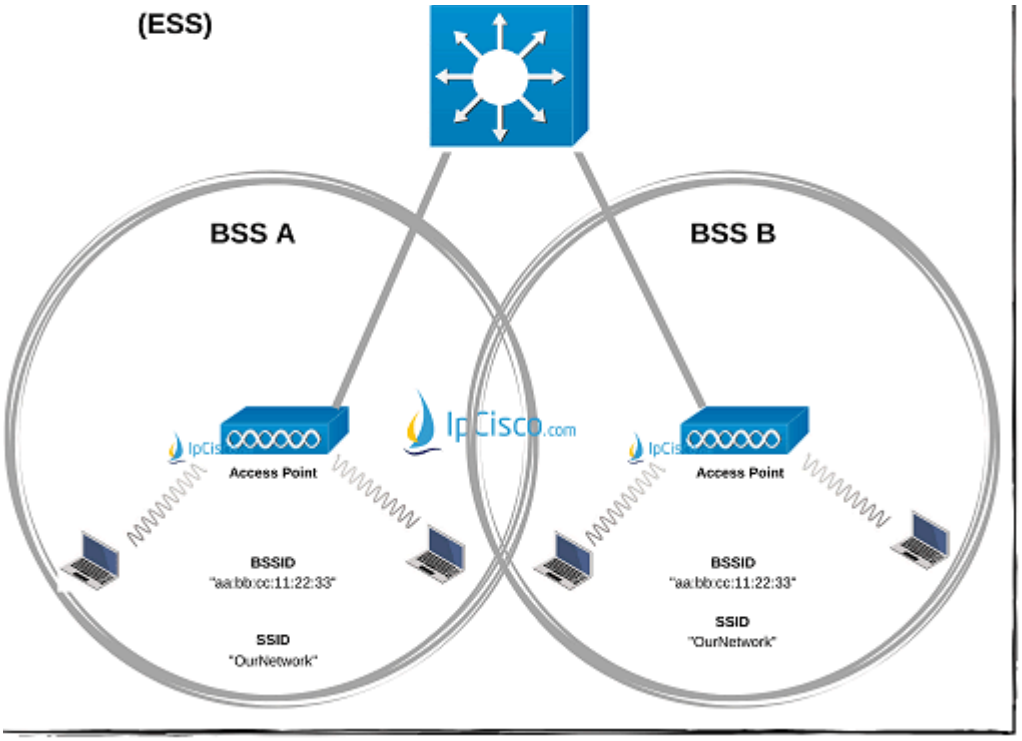
BSS
Access
Point
SSID
"OurNetwork"


(ESS)
BSS A
BSS B
Access Point
BSSID
"aa:bb:cc:11:22:33"
SSID
"OurNetwork"
Access Point
BSSID
"aa:bb:cc:11:22:33"
SSID
"OurNetwork"

1. Components of IEEE 802.11 Network

Every 802.11 network consists of four primary building blocks:

- **Station** (STA): Any device with a wireless network interface (NIC), such as a laptop or smartphone.
- **Access Point (AP**): A station that acts as a bridge between the wireless STAs and the wired network.
- **Basic Service Set (BSS)**: The smallest unit; a single AP and its associated STAs.
- **Extended Service Set (ESS**): Multiple BSSs connected via a wired backbone (Distribution System).
- **Distribution System** (**DS**): The logical component (usually Ethernet) that interconnects APs to allow for roaming.

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2. Physical Layer (PHY)

The Physical Layer handles the actual transmission of bits over the air. It is divided into two sub-layers:

- **PLCP** (Physical Layer Convergence Procedure): Prepares the MAC frame for transmission by adding a preamble and header.
- **PMD** (Physical Medium Dependent): Handles the actual modulation and coding of the signal (e.g., DSSS, OFDM, FHSS).
- **Key Standards**:
    - 802.11b: Uses DSSS (2.4 GHz).
    - 802.11a/g: Uses OFDM (5 GHz / 2.4 GHz).
    - 802.11n/ac: Introduced MIMO (Multiple Input Multiple Output).

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3. MAC Sub-Layers (DCF vs. PCF)

The MAC layer manages access to the shared medium to prevent collisions.

A. Distributed Coordination Function (DCF)

- **Mechanism**: Uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).

- **Contention-Based:** Stations "contend" for the channel. If the medium is busy, they wait a random "back-off" time.
- **Best Effort**: Used for standard data traffic like web browsing.

B. Point Coordination Function (PCF)

- **Mechanism**: Uses Polling.
- **Contention**-Free: The AP (acting as a Point Coordinator) "asks" each station if it has data to send.
- **Priority**: Used for time-sensitive traffic like Voice over IP (VoIP) or video streaming.

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4. Hidden and Exposed Station Problem

These are classic wireless challenges where physical distance causes communication failures.

Hidden Station Problem

- **Scenario**: Station A can see the AP. Station C can see the AP. But A and C cannot see each other.
- **Result**: A and C might send data to the AP at the same time, causing a collision at the AP.
- **Solution**: RTS/CTS (Request to Send / Clear to Send) handshake.

Exposed Station Problem

- **Scenario**: B is sending to A. C wants to send to D. C hears B's transmission and "thinks" the medium is busy, even though sending to D wouldn't interfere with A.
- **Result**: Unnecessary delay and wasted bandwidth.

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5. **Frame Format**

The 802.11 frame is more complex than an Ethernet frame because it must handle wireless-specific signaling.

- **Frame Control:** Contains protocol version, type (Management, Control, or Data), and power management bits.

- **Duration**/ID: Time reserved for the transmission.
- **Sequence Control**: Used for defragmentation and discarding duplicate frames.
- **Payload**: The actual data (up to 2312 bytes).
- **CRC**: Error checking.

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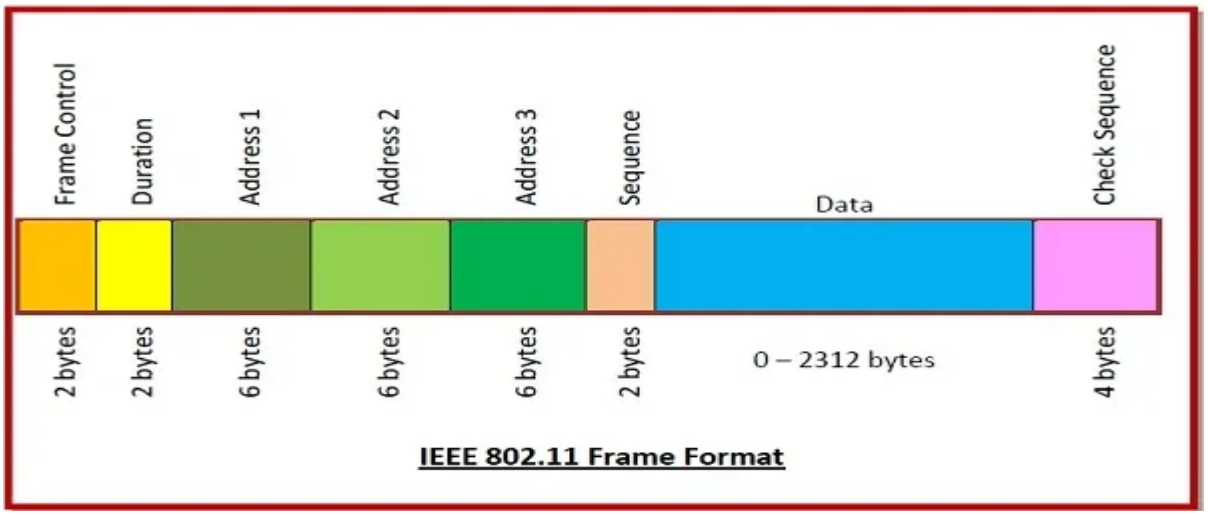
6. Addressing Mechanism

Unlike Ethernet (2 addresses), 802.11 uses up to 4 MAC addresses to manage the "hop" from the wireless station through the AP to the wired network.

- **Address** 1: Receiver Address (The immediate recipient).
- **Address** 2: Transmitter Address (The immediate sender).
- **Address** 3: Destination Address (The final destination, e.g., a router).
- **Address** 4: Source Address (The original sender, used in wireless bridge setups).



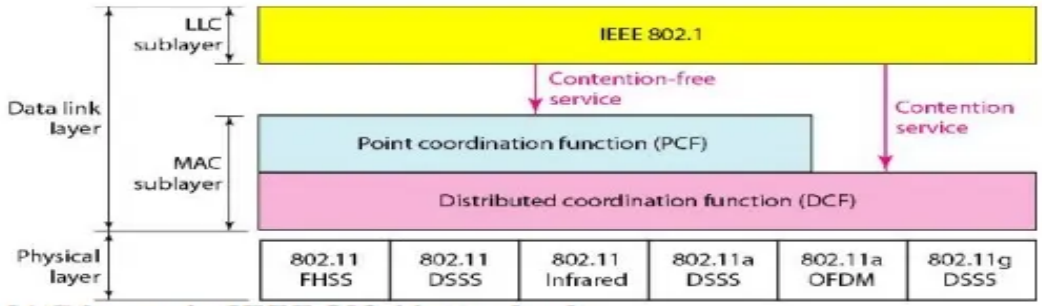


Figure 2.70 MAC layers in IEEE 802.11 standard

IEEE 802.15.1 Bluetooth
Bluetooth Layers (Protocol Stack)

Bluetooth does not follow the standard OSI model exactly; it has its own specific stack.

A. Physical Layer (Radio Layer)

- **Frequency**: Operates in the 2.4 GHz ISM band.
- **Frequency Hopping**: Uses FHSS (Frequency Hopping Spread Spectrum) at 1600 hops per second to reduce interference and increase security.
- **Modulation**: Uses GFSK (Gaussian Frequency Shift Keying).

B. Baseband Layer

- It is equivalent to the MAC sub-layer.
- Handles Connection Establishment, Channel Coding, and Timing.
- Manages two types of physical links:
    1. **SCO** (Synchronous Connection-Oriented): Used for real-time voice (fixed slots).
    2. **ACL** (Asynchronous Connection-Less): Used for data packets (packet switching).

C. L2CAP (Logical Link Control and Adaptation Protocol)

- Provides adaptation between upper-layer protocols and the Baseband.
- **Functions**: Segmentation and Reassembly of large packets, Multiplexing, and Quality of Service (QoS).

D. Middleware / Higher Layers

- RFCOMM: Provides a virtual serial port (emulates RS-232).
- SDP (Service Discovery Protocol): Allows devices to check what services (like printing or audio) other devices offer.
- OBEX: Used for exchanging objects like vCards or files.

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3. Operational States

A Bluetooth device transitions through several states to save power and maintain connectivity.

A. Major States:

1. **Standby**: The default low-power state where the device is not connected.
2. **Inquiry**: Used to discover other Bluetooth devices in the range.
3. **Page**: Used by the master to establish an actual connection with a specific device.
4. **Connected**: The device is actively part of a piconet.

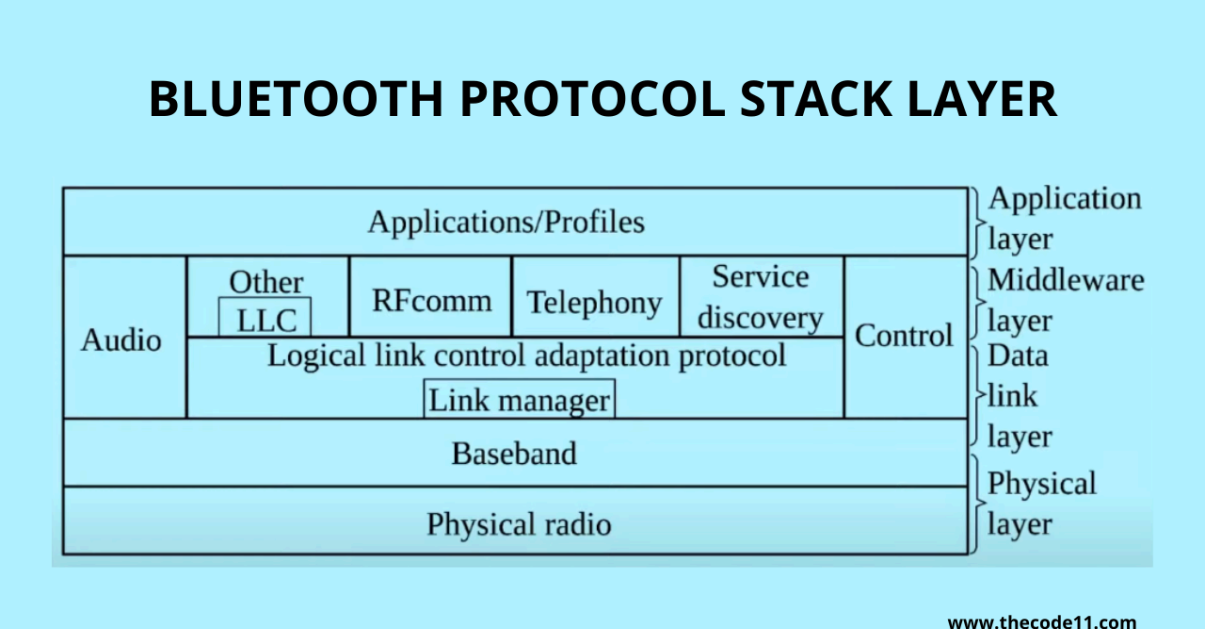
B. Power Saving Modes (Under Connected State):

For a 30-mark answer, you must detail these four sub-states:

- **Active Mode**: The device is actively transmitting or receiving data.
- **Sniff Mode**: The slave "listens" to the master at reduced intervals (e.g., every 100ms) to save battery.
- **Hold Mode**: The device stops data transmission for a specific duration but remains locked to the piconet.
- **Park Mode:** The device gives up its "Active Member Address," stays synchronized with the master's clock, but does not participate in traffic. This is the lowest power state.

4. Addressing and Security

- **BD_ADDR**: Every device has a unique 48-bit IEEE address.
- **Security**: Bluetooth uses Pairing (Authentication) and Encryption to ensure private communication between the Master and Slav



1. Introduction to IEEE 802.16 (WiMAX)

- **Definition:** IEEE 802.16 is the standard for **Wireless Metropolitan Area Networks (WMAN)**.
- **Range:** While Wi-Fi covers 100 meters, WiMAX can cover up to **50 km**.
- **Speed:** Provides data rates up to **70 Mbps**.
- **Usage:** It is used as a wireless alternative to cable and DSL for broadband access and for connecting Wi-Fi hotspots to the internet.

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2. WiMAX Architecture

The architecture is designed to handle point-to-multipoint communication over large distances.

A. Core Components

- **Subscriber Station (SS) / Mobile Station (MS):** The user's device (like a specialized modem or a smartphone).
- **Base Station (BS):** The central tower that provides connectivity to all Subscriber Stations within its range. It is connected to the Public Network (Internet) via high-speed fiber or microwave links.
- **Network Management System (NMS):** Manages the configuration and fault monitoring of the BS and SS.

B. Topology Models

1. **Point-to-Multipoint (PMP):** The Base Station serves multiple Subscriber Stations. All communication must pass through the BS.
2. **Mesh Topology:** Subscriber Stations can communicate with each other, acting as routers to pass data along to the Base Station. This extends the range significantly.

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3. WiMAX Layers (Protocol Stack)

WiMAX is unique because it splits the Data Link Layer into three specific sub-layers to handle high-speed data and Quality of Service (QoS).

A. Physical Layer (PHY)

- **Modulation:** Uses **OFDM** (Orthogonal Frequency Division Multiplexing) to handle non-line-of-sight (NLOS) signals and interference.
- **Duplexing:** Supports both **TDD** (Time Division Duplex) and **FDD** (Frequency Division Duplex).
- **Adaptive Modulation:** The BS can change the modulation (e.g., from 64-QAM to QPSK) depending on the signal strength and distance of the subscriber.

B. MAC Layer (Media Access Control)

The MAC layer is divided into three sub-layers:

1. **Service Specific Convergence Sub-layer (CS):** Receives data from higher-level protocols (like IP or Ethernet) and converts them into a format the MAC layer understands.
2. **MAC Common Part Sub-layer (CPS):** This is the "brain." It handles **Bandwidth Allocation**, **Connection Establishment**, and **Scheduling**. Unlike Wi-Fi (which uses contention), WiMAX uses a **scheduling algorithm** where the BS decides exactly when a station can transmit.
3. **Security Sub-layer:** Handles encryption (RSA/AES), authentication, and key exchange to ensure data privacy.

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4. WiMAX Services (Quality of Service - QoS)

For a 30-mark answer, you **must** list these 5 service classes. WiMAX is designed to prioritize different types of data.

- **UGS (Unsolicited Grant Service):** For real-time data with fixed rates, like **T1/E1 lines** or VoIP without silence suppression.
- **rtPS (Real-time Polling Service):** For variable-sized data packets that must be sent regularly, such as **MPEG video streaming**.
- **ertPS (Extended Real-time Polling Service):** A hybrid used for **VoIP with silence suppression** (saves bandwidth when no one is talking).
- **nrtPS (Non-real-time Polling Service):** For high-bandwidth applications that are not time-sensitive, like **large file transfers (FTP)**.
- **BE (Best Effort):** The lowest priority service, used for standard **Web Browsing**.

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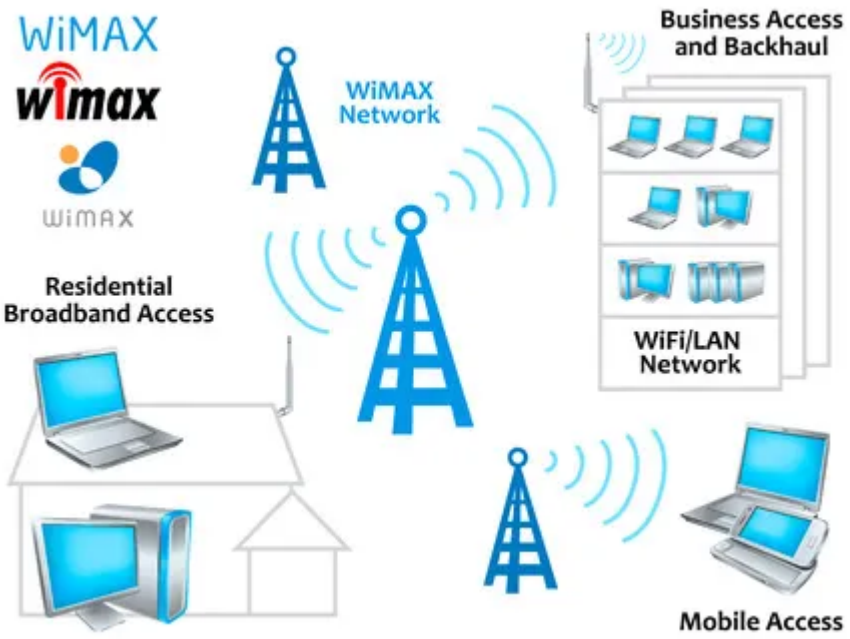
5. Comparison & Conclusion

- **WiMAX vs. Wi-Fi:** Mention that WiMAX is for cities (WMAN) while Wi-Fi is for buildings (WLAN). WiMAX uses a scheduled MAC, while Wi-Fi uses a contention-based MAC (CSMA/CA).
- **Conclusion:** WiMAX serves as a vital "last-mile" solution, especially in rural areas where laying fiber optic cables is too expensive.



Comparison between Bluetooth, IEEE 802.11, and IEEE 802.16

Feature	Bluetooth	IEEE 802.11 (Wi-Fi)	IEEE 802.16 (WiMAX)
IEEE Standard	802.15.1	802.11 (a/b/g/n/ac/ad)	802.16 (d/e/m)
Technology Type	Wireless Personal Area Network (WPAN)	Wireless Local Area Network (WLAN)	Wireless Metropolitan Are (WMAN)
Frequency Band	2.4 GHz ISM band	2.4 GHz and 5 GHz	2–11 GHz (commonly 2.5,
Typical Range	10 m (up to 100 m for Class 1)	30–100 m indoors	7–50 km
Data Rate	Up to 3 Mbps (Classic), 1 Mbps (BLE)	Up to 600 Mbps (802.11n), multi-Gbps (802.11ac/ad)	Up to 100–150 Mbps
Power Consumption	Very low (especially BLE)	High	Moderate to high
Primary Use Case	Cable replacement, peripheral connectivity	Internet access, LAN connectivity	Last-mile broadband, rural connectivity
Network Size	Up to 7 active devices (piconet)	Up to 255 devices per AP	Scalable to hundreds of CPEs
QoS Support	Limited (basic prioritization)	Supported via 802.11e	Strong QoS with UGS, rtPS, nrtPS, BE
Encryption	AES (modern versions)	WPA2/WPA3 (AES-based)	DES (mandatory), AES (optional)
Modulation & Resilience	FHSS (Frequency Hopping Spread Spectrum)	DSSS, OFDM	OFDM, scalable OFDMA
Interference with 2.4 GHz Devices	Yes (with Wi-Fi) but mitigated by FHSS	Yes (with Bluetooth)	Minimal (different bands)

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Detailed Comp
Summary and

Detailed Comparison by Feature

1. Range and Coverage

- **Bluetooth:** Designed for personal space (10 m typical), suitable for device-to-device links.
- **Wi-Fi (**802.11): Covers rooms or buildings (30–100 m), ideal for home/office networks.
- **WiMAX** (802.16): Extends to 50 km, enabling city-wide or rural broadband without wired infrastructure.

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2. Data Throughput

- **Bluetooth**: Low to moderate (1–3 Mbps), sufficient for audio and control signals.
- **Wi-Fi**: High throughput (up to multi-Gbps with 802.11ac/ad), supports HD video streaming and large file transfers.

- **WiMAX**: Moderate to high (up to 150 Mbps), suitable for fixed broadband delivery.

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3. Power Efficiency

- **Bluetooth** (especially BLE): Extremely power-efficient, enabling battery operation for months or years.
- **Wi-Fi**: High power draw, limiting use in battery-powered IoT devices.
- **WiMAX**: Moderate to high power, typically used in fixed installations with external power.

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4. Network Topology and Scalability

- **Bluetooth**: Piconet (master-slave, up to 7 active devices), limited scalability.
- **Wi-Fi**: Star topology with access points; supports dozens of clients per AP.
- **WiMAX**: Point-to-multipoint, highly scalable for hundreds of customer premises equipment (CPEs).

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5. Interference and Coexistence

- Both Bluetooth and Wi-Fi operate in the 2.4 GHz ISM band, leading to potential interference. However, Bluetooth's Frequency Hopping Spread Spectrum (FHSS) helps mitigate this.
- WiMAX operates in licensed or less congested bands (e.g., 3.5 GHz), reducing interference risks.
- Some Bluetooth versions (e.g., Bluetooth 3.0 + HS) use 802.11 for high-speed data transfer, coordinating between the two protocols